# MSG No. 47.251 $Pm'mm$ [ Type III, orthorhombic ]

Table 1: Wyckoff site: `1a`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,2,3,4,5,6,7,8]` |

Table 2: Wyckoff site: `1b`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 3: Wyckoff site: `1c`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 4: Wyckoff site: `1d`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 5: Wyckoff site: `1e`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, 0\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 6: Wyckoff site: `1f`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 7: Wyckoff site: `1g`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 8: Wyckoff site: `1h`, site symmetry: `m'mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |

Table 9: Wyckoff site: `2i`, site symmetry: `2mm`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,2,3,4]` |
| 2 | $[-x,\ 0,\ 0]$ | `[5,6,7,8]` |

Table 10: Wyckoff site: `2j`, site symmetry: `2mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[1,2,3,4]` |
| 2 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[5,6,7,8]` |

Table 11: Wyckoff site: `2k`, site symmetry: `2mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[1,2,3,4]` |
| 2 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[5,6,7,8]` |

Table 12: Wyckoff site: `2l`, site symmetry: `2mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[1,2,3,4]` |
| 2 | $\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[5,6,7,8]` |

Table 13: Wyckoff site: `2m`, site symmetry: `m'2'm`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ 0]$ | `[1,4,5,8]` |
| 2 | $[0,\ -y,\ 0]$ | `[2,3,6,7]` |

Table 14: Wyckoff site: `2n`, site symmetry: `m'2'm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{2}\right]$ | `[1,4,5,8]` |
| 2 | $\left[0,\ -y,\ \frac{1}{2}\right]$ | `[2,3,6,7]` |

Table 15: Wyckoff site: `2o`, site symmetry: `m'2'm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ y,\ 0\right]$ | `[1,4,5,8]` |
| 2 | $\left[\frac{1}{2},\ -y,\ 0\right]$ | `[2,3,6,7]` |

Table 16: Wyckoff site: 2p, site symmetry: m'2'm

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, y, \frac{1}{2}\right]$ | [1,4,5,8] |
| 2 | $\left[\frac{1}{2}, -y, \frac{1}{2}\right]$ | [2,3,6,7] |

Table 17: Wyckoff site: 2q, site symmetry: m'm2'

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | [1,3,6,8] |
| 2 | $[0, 0, -z]$ | [2,4,5,7] |

Table 18: Wyckoff site: 2r, site symmetry: m'm2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, z\right]$ | [1,3,6,8] |
| 2 | $\left[0, \frac{1}{2}, -z\right]$ | [2,4,5,7] |

Table 19: Wyckoff site: 2s, site symmetry: m'm2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, z\right]$ | [1,3,6,8] |
| 2 | $\left[\frac{1}{2}, 0, -z\right]$ | [2,4,5,7] |

Table 20: Wyckoff site: 2t, site symmetry: m'm2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | [1,3,6,8] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, -z\right]$ | [2,4,5,7] |

Table 21: Wyckoff site: 4u, site symmetry: m'..

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, z]$ | [1,8] |
| 2 | $[0, -y, -z]$ | [2,7] |
| 3 | $[0, -y, z]$ | [3,6] |
| 4 | $[0, y, -z]$ | [4,5] |

Table 22: Wyckoff site: 4v, site symmetry: m'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, y, z\right]$ | [1,8] |
| 2 | $\left[\frac{1}{2}, -y, -z\right]$ | [2,7] |

*continued ...*

Table 22

| No. | position | mapping |
|---|---|---|
| 3 | $\left[\frac{1}{2}, -y, z\right]$ | `[3,6]` |
| 4 | $\left[\frac{1}{2}, y, -z\right]$ | `[4,5]` |

Table 23: Wyckoff site: `4w`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, z]$ | `[1,3]` |
| 2 | $[x, 0, -z]$ | `[2,4]` |
| 3 | $[-x, 0, -z]$ | `[5,7]` |
| 4 | $[-x, 0, z]$ | `[6,8]` |

Table 24: Wyckoff site: `4x`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{2}, z\right]$ | `[1,3]` |
| 2 | $\left[x, \frac{1}{2}, -z\right]$ | `[2,4]` |
| 3 | $\left[-x, \frac{1}{2}, -z\right]$ | `[5,7]` |
| 4 | $\left[-x, \frac{1}{2}, z\right]$ | `[6,8]` |

Table 25: Wyckoff site: `4y`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, 0]$ | `[1,4]` |
| 2 | $[x, -y, 0]$ | `[2,3]` |
| 3 | $[-x, y, 0]$ | `[5,8]` |
| 4 | $[-x, -y, 0]$ | `[6,7]` |

Table 26: Wyckoff site: `4z`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, y, \frac{1}{2}\right]$ | `[1,4]` |
| 2 | $\left[x, -y, \frac{1}{2}\right]$ | `[2,3]` |
| 3 | $\left[-x, y, \frac{1}{2}\right]$ | `[5,8]` |
| 4 | $\left[-x, -y, \frac{1}{2}\right]$ | `[6,7]` |

Table 27: Wyckoff site: `8A`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | `[1]` |
| 2 | $[x, -y, -z]$ | `[2]` |
| 3 | $[x, -y, z]$ | `[3]` |

*continued ...*

Table 27

| No. | position | mapping |
|---|---|---|
| 4 | $[x, y, -z]$ | `[4]` |
| 5 | $[-x, y, -z]$ | `[5]` |
| 6 | $[-x, -y, z]$ | `[6]` |
| 7 | $[-x, -y, -z]$ | `[7]` |
| 8 | $[-x, y, z]$ | `[8]` |